



ENGINEER 4A03
Ethics, Equity and Law in Engineering
Undergraduate Studies
Winter 2025
Course Outline

Current as of Sun 2025-01-05 13:44:23; see the A2L for the most up-to-date version of this document

CALENDAR/COURSE DESCRIPTION

General introduction to the ethical, equitable and legal aspects of engineering practice. Topics include the engineering act and code of ethics, employment equity, legal duties and liabilities of the professional engineer, contracts, and the tort of negligence.
Three lectures; both terms

PRE-REQUISITES AND ANTI-REQUISITES

Prerequisite(s): Registration in Level IV or above of any Engineering program or permission of the instructor. Not open to Engineering & Society program.
Antirequisite(s): ENGPYHS 2S03, 4C03

INSTRUCTOR OFFICE HOURS AND CONTACT INFORMATION

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Office Hours:
TBA

TEACHING ASSISTANT OFFICE HOURS AND CONTACT INFORMATION

See A2L for TA info

COURSE WEBSITE/ALTERNATE METHODS OF COMMUNICATION

The primary method of communication will be

1. Avenue To Learn (A2L, <http://avenue.mcmaster.ca/>) news postings for announcements - make sure to set your A2L email settings so it emails these to you.
2. MS Teams ("Course Forum") for announcements, questions, etc.
3. Email for individual messages (Try to use the Teams Forum to ask questions that others might want the answer to as well!)

COURSE INTENDED LEARNING OUTCOMES

Upon successful completion of the course, you will be able to:

1. Explain professional engineering licensing and how it protects the public
2. Analyze and make ethical, equitable, and legal arguments in a professional engineering context

MATERIALS AND FEES

REFERENCE TEXTS:

- Required: Ontario Professional Engineers [Act](#) & Regulation [941](#)
- [Optional] Practical Law of Architecture, Engineering, and Geoscience, by Brian Samuels and Doug Sanders
- [Optional] Law for Professional Engineers by Don Marston
- [Optional] Canadian Professional Engineering and Geoscience: Practice and Ethics, by Gordon Andrews, Patricia Shaw, and John McPhee

COURSE FORMAT AND EXPECTATIONS

This course has lectures via Teams, quizzes for each week via A2L, followed by a final exam (derived from the same content as the quizzes).

Attend the classes (or watch the recordings), study the content, and complete the quizzes to increase your success!

ACCREDITATION LEARNING OUTCOMES
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In pursuit of the course's ILOs, this course will further impart the following CEAB graduate attributes, and specifically measure your success at these via the following associated indicators:

08 – PROFESSIONALISM

An understanding of the roles and responsibilities of the professional engineer in society, especially the primary role of protection of the public and the public interest.

8.1 Describes the duty of a Professional Engineer to the public, client, employer, and the profession.

8.2 Integrates appropriate standards, codes, legal and regulatory factors into decision making.

09 - IMPACT OF ENGINEERING ON SOCIETY AND THE ENVIRONMENT

An ability to analyze societal and environmental aspects of engineering activities. Such ability includes an understanding of the interactions that engineering has with the economic, health, safety, legal, and cultural aspects of society, the uncertainties in the prediction of such interactions; and the concepts of sustainable design and development and environmental stewardship.

9.1 Evaluates the environmental impact of engineering activities, identifies uncertainties in decisions, and promotes sustainable design.

9.2 Evaluates the social impact of engineering activities, including health, safety, legal, cultural, and other relevant factors, and identifies uncertainties in decisions.

10 - ETHICS AND EQUITY

An ability to apply professional ethics, accountability, and equity.

10.1 Applies ethical frameworks and reasoning, including in situations where there are possible conflicting interests among the stakeholders.

10.2 Applies the principles of equity and universal design to ensure equitable treatment of all stakeholders.

11 - ECONOMICS AND PROJECT MANAGEMENT

An ability to appropriately incorporate economics and business practices including project, risk, and change management into the practice of engineering and to understand their limitations.

11.3 Identifies, characterizes, assesses, and manages risks to project success.

12 - LIFE-LONG LEARNING

An ability to identify and to address their own educational needs in a changing world in ways sufficient to maintain their competence and to allow them to contribute to the advancement of knowledge

12.1 Reflects on one's own educational needs and opportunities for growth.

12.2 Seeks and acquires appropriate external information as required, including showing awareness of sources of information and ability to critically evaluate them.

For more information on Accreditation, please visit: <https://www.engineerscanada.ca>

COURSE SCHEDULE

Week-by-Week 4A03 Topics, Sub-ILOs, and Associated CEAB Indicators

1. Course intro, and intro to Engineering as a Self-Regulated Profession:
 - a. Scope of Engineering
 - b. Engineering as a self-regulating profession
 - i. What is a profession?
 - ii. What is a self-regulating profession?
 - iii. What is PEO and what does it do?
 - c. Enforcement: (8.2)
 - i. Title & practice offenses.
 - ii. When do you need a license?
 - iii. What's a CofA and when do you need it?
2. Ethics 1: Classical Morality & Ethics
 - a. Critical Thinking – Evaluating Arguments
 - i. Structure of an argument
 - ii. Argument validity
 - iii. Logical fallacies
 - b. Classical Ethical Frameworks: What do these each say about how to determine what is morally right, and what are some challenges to them?
 - i. Divine Command Theory
 - ii. Virtue Ethics
 - iii. Utilitarianism
 - iv. Duty Ethics
 - v. Contractarianism
 - c. How to Reason Ethically in Professional Engineering Contexts
 - i. Stages of Moral Development
 - ii. That the professional code of ethics determines right and wrong in engineering contexts.
3. Ethics 2: Understanding what the codes say and how they're interpreted.
 - a. Overview of The Code of Ethics and Definition of Professional Misconduct
 - i. Why are there two documents? What does each mean?
 - ii. Complaints & discipline process brief overview to put Defn of PM and CoE into context: Who assesses whether you've committed professional misconduct and why does this matter for interpreting it?
 - iii. What are the consequences of committing PM or breaking the CoE?
 - b. Deep dive into sections of PM and CoE:
 - i. What does each section mean?
 - ii. Who are the parties involved?
 1. The Public

- 2. Employers
 - 3. Clients
 - 4. Practitioners
- iii. Who could this section be an obligation to? (8.1). Questions like “do you have obligations to your employer here?”
- 4. Ethics 3: Resolving Ethical Dilemmas
 - a. Applying CoE and PM in resolving ethical dilemmas (8.2 use of codes). (10.1 applies ethical frameworks and reasoning) Questions like “what should you do to resolve this dilemma?”
 - i. Resolving environmental ethical dilemmas including employer breaking environmental law (9.1)
 - ii. Recognizing need for engineers to include environmental costs in assessing economic viability of a project (9.1)
 - iii. Recognizing need for engineers to include social costs in assessing economic viability of a project (9.2)
 - iv. Whistleblowing; how duty to employer factors in
 - v. Examples with conflicts of interest
- 5. Ethics 4: CoE & PM Special topics & Examples:
 - a. Equity requirements via PM items d & item n:
 - i. Ontario human rights code (also 10.2 – resolving dilemmas recognizing Ontario HRC violations)
 - 1. resolving dilemmas expecting equity
 - 2. Employment standards act
 - ii. Harassment from an equity lens (10.2) What could interfere with an engineering relationship? What does vexation mean?
 - b. The Engineer's Seal
 - i. What it's for and when to use it
 - ii. How to use it – other information to provide
 - c. Other information that impacts PM via item g:
 - i. Use of an engineering Seal
 - ii. Advertising standards
 - iii. CofA insurance requirements
 - iv. Consulting engineers
- 6. Ethics 5: Engineering Complaints and Discipline:
 - a. Complaints and discipline process details:
 - i. Stages of the complaint & discipline process
 - ii. What does the Complaints Committee do?
 - iii. What does the Discipline Committee do?
 - iv. What are the powers of the discipline committee?
 - v. Case studies: examples from engineering dimensions
 - b. Professional responsibility and need to seek external information (12.2)
 - i. Would you need to be aware of codes and standards not set by the government (like a technical society)?
 - ii. “The expert assesses that the engineer didn't use the code they obviously ought to have used knowing it was for manufacturing & warehousing.” – if no one ever told the engineer there were different codes for different types of buildings could this still be negligence?
- 7. Law 1: Canadian Common Law and Intro to Tort

- a. Common law background information:
 - i. Brief intro to the Canadian legal system, constitution, and charter of rights and freedoms
 - ii. Brief intro to common law, and public vs. private law; Degree of proof needed in public vs. private law.
 - iii. Brief intro to precedent; and trial and appeals process
 - b. Tort intro
 - i. how suing works - Unit Farm
 1. Appreciating social impact of engineering through appreciating implications of giving engineering advice – how informal engineering advice (engineering activity) can be taken seriously and raise liability. (9.2)
 - ii. Principals & purpose of tort law
8. Law 2: Tort Law Details (Integrating Legal & Regulatory factors into decision making (8.2))
 - a. Development of tort law: Donahue v. Stevens, Hedley Byrne, Wolverine Tube
 - b. Damages in Tort
 - i. Direct vs. Indirect vs. Liquidated Damages
 - c. Disclaimers – when are they effective and not effective? Can you use a disclaimer to get rid of your engineering responsibilities?
 - d. Examples of torts other than negligence in professional service
9. Law 3: Contract Law (Integrating Legal & Regulatory factors into decision making (8.2))
 - a. What is a Contract?
 - i. The essential elements of a contract
 - ii. Verbal contracts possible? Burden of proof and the statute of frauds.
 - b. Damages in Contract
 - i. Direct vs. Indirect vs. Liquidated Damages
 - ii. The Hadley vs. Baxendale Principle
 - c. Duty of honesty; Misrepresentation
 - d. Discharge of Contracts
 - e. Importance of consideration: Gratuitous Promise and Equitable Estoppel
10. Law 4: Contract Special Topics and Risk Management
 - a. Offer & acceptance; consideration, and right to revoke: how tendering works
 - b. Unilateral mistake in bid submission
 - c. Risk Management (11.3)
 - i. Steps in risk management
 - ii. Hazard analysis
 - iii. Risk management in contract via liquidated damages provisions, force majeure provisions, and disclaimers
11. Law 5: Special Areas of Law
 - a. Ontario limitations act – How long can you take to sue someone?
 - b. Alternative dispute resolution and advantages for international disagreements
 - c. Privacy law
 - d. Products liability; difference in Canada & US, strict vs. absolute vs. normal liability.
12. Review
13. Review

ASSESSMENT

Assessment Item	Each	Number	Total
Quizzes	4%	11	44%
Final Exam	56%	1	56%
Grand Total			100%

Note that the MSF policy is as follows: MSF'd quizzes will be given an extension. Quizzes are normally due 1 week after the lecture they cover, but if for whatever reason you need an extension you can have as long as I can give you: there is no late penalty until I pull the final quiz marks from A2L one day after the final exam - any quizzes still not done by then will not be able to receive marks.

FINAL EXAM

1. The final exam is a closed book written exam answered on paper, but does allow you to bring a double-sided crib sheet
2. Since the exam will have questions that are the same or very similar to the quizzes, be sure to put anything that you needed to answer the quizzes onto your crib sheet.

EQUITY, DIVERSITY, AND INCLUSION

Every registered student belongs in this course. Diversity of backgrounds and experiences is expected and welcome. You can expect your Instructor to be respectful of this diversity in all aspects of the course, and the same is expected of you.

The Department of Engineering Physics is committed to creating an environment in which students of all genders, cultures, ethnicities, races, sexual orientations, abilities, and socioeconomic backgrounds have equal access to education and are welcomed and treated fairly. If you have any concerns regarding inclusion in our Department, in particular if you or one of your peers is experiencing harassment or discrimination, you are encouraged to contact the Chair, Associate Undergraduate Chair, Academic Advisor or to contact the [Equity and Inclusion Office](#).

PHYSICAL AND MENTAL HEALTH

For a list of McMaster University's resources, please refer to the [Student Wellness Centre](#).

ACADEMIC INTEGRITY

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the [Academic Integrity Policy](https://secretariat.mcmaster.ca/university-policies-procedures-guidelines/), located at <https://secretariat.mcmaster.ca/university-policies-procedures-guidelines/>



The following illustrates only three forms of academic dishonesty:

1. plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
2. improper collaboration in group work.
3. copying or using unauthorized aids in tests and examinations.

AUTHENTICITY / PLAGIARISM DETECTION

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. **All submitted work is subject to normal verification that standards of academic integrity have been upheld** (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

COURSES WITH AN ON-LINE ELEMENT

McMaster is committed to an inclusive and respectful community. These principles and expectations extend to online activities including electronic chat groups, video calls and other learning platforms.

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn (A2L), LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure, please discuss this with the course instructor.

CONDUCT EXPECTATIONS

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the [Code of Student Rights & Responsibilities](#) (the "Code"). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online**.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students' access to these platforms.

ACADEMIC ACCOMMODATION OF STUDENTS WITH DISABILITIES

Students with disabilities who require academic accommodation must contact [Student Accessibility Services](#) (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's [Academic Accommodation of Students with Disabilities](#) policy.

COURSE POLICY ON MISSED WORK, EXTENSIONS, AND LATE PENALTIES

It is your responsibility to regularly check both the course Teams page and A2L for updates and announcements. Policies for missed and late work are explained in the Assessment section.

SUBMISSION OF REQUEST FOR RELIEF FOR MISSED ACADEMIC WORK

In the event of an absence for medical or other reasons, students should review and follow the Academic Regulation in the Undergraduate Calendar "Requests for Relief for Missed Academic Term Work".

1. Relief for missed academic work worth less than 25% of the final grade resulting from medical or personal situations lasting up to three calendar days:
 - Use the [McMaster Student Absence Form](#) (MSAF) on-line self-reporting tool. No further documentation is required.
 - Students may submit requests for relief using the MSAF once per term.
 - An automated email will be sent to the course instructor, who will determine the appropriate relief. Students must immediately follow up with their instructors. Failure to do so may negate the opportunity for relief.
 - a. **Normal MSAF relief policy for this course is covered in the assessment section**
 - The MSAF cannot be used to meet a religious obligation or to celebrate an important religious holiday.
 - The MSAF cannot be used for academic work that has already been completed or attempted.
 - An MSAF applies only to work that is due within the period for which the MSAF applies, i.e. the 3-day period that is specified in the MSAF; however, all work due in that period can be covered by one MSAF.
 - The MSAF cannot be used to apply for relief for any final examination or its equivalent. See *Petitions for Special Consideration* above.
2. For medical or personal situations lasting more than three calendar days, and/or for missed academic work worth 25% or more of the final grade, and/or for any request for relief in a term where the MSAF has been used previously in that term:
 - Students must report to their Faculty Office to discuss their situation and will be required to provide appropriate **supporting documentation**.
 - If warranted, the Faculty Office will approve the absence, and the instructor will determine appropriate relief.

ACADEMIC ACCOMMODATION FOR RELIGIOUS, INDIGENOUS OR SPIRITUAL OBSERVANCES (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the [RISO](#) policy. Students should submit their request to their Faculty Office **normally within 10 working days** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

COPYRIGHT AND RECORDING

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

EXTREME CIRCUMSTANCES

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, A2L and/or McMaster email.